Corporate-Owned (COD) & BYOD Device Management

Implementation Solution Guide

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Executive Summary — Corporate-Owned (COD) & BYOD Device Management

Most organizations run **two** device realities at once: **corporate-owned devices (COD)** that IT buys and fully controls, and **personal (BYOD)** devices employees already own and want to use for work. Managed separately, they create duplicate policies, inconsistent security, and confused users.

This solution unifies both under **Microsoft Intune** and **one Microsoft Entra identity**: the same user signs in with the same corporate account on **both** their company device and their personal device, while IT applies the **right level of control to each** based on who owns it.

• **Corporate-owned (COD)** → full MDM enrollment, device compliance, and configuration. IT owns the device end to end.

• **Personal / BYOD** → lighter-touch management: **App Protection (MAM)** that secures corporate data inside apps without taking over the employee's device — or optional personal MDM enrollment.

• **One identity, both devices** → the user's single Entra account works everywhere; **ownership-based targeting **(device filters + assignments) routes the correct policy to each device automatically, and** Conditional Access** grants access from a compliant company device *or* a protected app on a personal one.

Built the standard repeatable way — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — with exact, click-by-click Intune steps, a requirements guide that tells you how to verify *and* fix each prerequisite, five ready-to-run PowerShell scripts (Microsoft Graph), and testing/troubleshooting guides with clear pass/fail gates.

Outcome: a single, coherent device-management model where one employee account works on both a corporate and a personal device, corporate data stays protected on each, and access is governed by compliance and app protection — delivered by your own team without a consulting engagement.

1. Solution Overview — Corporate-Owned (COD) & BYOD Device Management

What this solution is

A unified way to manage **two device-ownership models** in Microsoft Intune under **one Microsoft Entra identity**:

1. **Corporate-Owned Devices (COD)** — devices the organization buys and owns. They are fully enrolled in Intune (MDM), receive configuration and compliance policies, and are managed end to end by IT.

1. **Bring Your Own Device (BYOD)** — personal devices the employee owns. The corporate **identity and data **are managed without taking over the personal device, primarily through** App Protection Policies (MAM)**, with optional lightweight personal MDM enrollment.

The defining design goal of this solution: **the same user account signs in on both device types.** A user authenticates with their single Entra ID everywhere; Intune applies the correct controls to each device based on **ownership**, not on a second account.

How it works

1. **One identity.** Each user has a single Microsoft Entra account. That account is what enrolls the corporate device and what signs into work apps on the personal device — there is no separate "BYOD account."

1. **Ownership is tagged.** Every enrolled device is marked **Corporate** or **Personal** (by enrollment method, corporate device identifiers, or enrollment-restriction defaults).

1. **Ownership-based targeting.** Policies are assigned to the **user** (or a user group), then scoped with **device filters** on `deviceOwnership` so:

- **Corporate** devices receive full **device compliance** + **configuration** policies.
- **Personal** devices receive **App Protection (MAM)** policies that encrypt corporate data, enforce a PIN, and block copy/paste or "Save As" to personal locations.

1. **Conditional Access ties it together.** A single Conditional Access policy for the user grants access when the request comes from a **compliant corporate device** *or* from an **approved app with an app-protection policy** on a personal device — so the one account works securely from both.

Management models (choose per ownership)

- **COD — full MDM:** complete device management, configuration profiles, compliance, and (optionally) Autopilot/ADE/Android Enterprise corporate enrollment for zero-touch provisioning.
- **BYOD — App Protection / MAM (recommended):** secures corporate data *inside* managed apps with no full device enrollment; least intrusive for employees, strong data-leak protection.
- **BYOD — personal MDM enrollment (optional):** when a personal device must receive a few device settings; still tagged Personal and scoped accordingly.

This solution standardizes on **full MDM for COD** and **App Protection for BYOD**, with personal MDM called out as an option.

Benefits

- **One account, both devices** — no duplicate identities; users sign in with what they already know.
- **Right-sized control** — full management where IT owns the device; data-only protection where the employee does.
- **Lower friction BYOD** — employees keep their device private; corporate data stays contained.
- **Consistent security** — compliance for COD, app protection for BYOD, unified by Conditional Access.

• **Scales cleanly** — ownership-based targeting means you assign once to the user and the right policy follows the device.

Limitations (be honest with buyers)

• **App Protection (MAM)** secures data **inside supported apps** (Microsoft 365 and SDK-enabled apps); it does not manage the whole personal device or unmanaged apps.

• BYOD users must use the **managed/approved apps** (e.g. Outlook, Edge, Teams) to get corporate data;

built-in personal mail/browser apps are blocked from corporate data by design.

• COD zero-touch provisioning (Autopilot/ADE/Android Enterprise) is referenced but each platform's full enrollment is covered by its own solution; this solution unifies the **model**.

• A user signing in on a brand-new device enrolls/registers that device the first time.

The standard deployment model

This solution uses **one** model so the process is repeatable across our catalog:

> ***One Microsoft Entra identity per user; corporate-owned devices under full MDM and personal devices under App Protection (MAM); ownership-based targeting routes the right policy to each; a single Conditional Access policy governs access from both.***

Every step follows the repeatable pattern: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

2. Requirements — Corporate-Owned (COD) & BYOD Device Management

Confirm **every** requirement below before you begin. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.** Category-wide prerequisites (tenant, Intune licensing, admin roles, pilot group, supported OS, network) are covered once in the **Global Requirements — Intune Solutions** document; the items here are specific to running COD and BYOD under one identity.

R1. A single Microsoft Entra identity per user

Description: Each user has exactly one Entra ID account used for both device types. **Why:** The whole design depends on one account signing in on both the corporate and the personal device; two accounts would split policy and defeat the goal. **Verify:** Entra → **Users**; the pilot user exists once with a licensed account. **Implement:** Consolidate duplicate accounts; ensure the user signs in with the same UPN everywhere.

R2. Intune licensing that includes App Protection (MAM)

Description: A license plan that covers both device management (MDM) and app management (MAM). **Why:** COD uses MDM; BYOD uses App Protection (MAM). Both must be licensed for the same user.

Verify: Entra → user → **Licenses**; an Intune-bearing plan (M365 E3/E5, Business Premium, or Intune Plan 1) is assigned. **Implement:** Assign the appropriate Microsoft 365/Intune license to all pilot users.

R3. Microsoft Entra ID P1 (for Conditional Access)

Description: Entra ID P1 (or higher) for Conditional Access. **Why:** A single Conditional Access policy is what lets one account access work from a compliant COD *or* an app-protected BYOD device; CA requires P1. **Verify:** Entra → **Conditional Access** opens and policies can be created (P1 present). **Implement:** Assign Entra ID P1 (included in M365 E3/E5/Business Premium) to pilot users.

R4. Enrollment restrictions configured for both ownership types

Description: Device platform/enrollment restrictions that **allow personal (BYOD)** enrollment for the BYOD path and treat the correct platforms as **corporate** for COD. **Why:** A restriction that blocks personal enrollment, or that auto-marks all devices corporate, breaks the ownership distinction this solution relies on. **Verify:** Intune → **Devices** → **Enrollment** → **Enrollment device platform restrictions**; the pilot restriction shows the intended **Personal = Allow/Block** per platform. **Implement:** Edit the default or a pilot-targeted restriction so BYOD platforms allow personal, and keep COD platforms enrolling as corporate (see R5).

R5. A way to mark devices Corporate

Description: Corporate device identifiers and/or corporate enrollment methods so COD devices are tagged **Corporate**. **Why:** Ownership = Corporate is the signal that routes full MDM policy (vs. App Protection) to a device. **Verify:** Intune → **Devices** → **Enrollment** → **Corporate device identifiers** (serials/IMEIs), or a device's **Ownership** field reads **Corporate**. **Implement:** Add corporate identifiers, use corporate enrollment programs (Autopilot/ADE/Android Enterprise corporate), or set **Ownership = Corporate** on the device record.

R6. Company Portal app

Description: The Intune **Company Portal** available on managed devices. **Why:** Company Portal performs MDM enrollment for COD and is the broker that registers a personal device and surfaces app-protection-enabled apps for BYOD. **Verify:** Intune → **Apps**; Company Portal is added for the relevant platforms and assigned to the pilot group. **Implement:** Add Company Portal per platform and assign it to the pilot group.

R7. App Protection (MAM) capability and approved apps

Description: App Protection Policies plus the approved managed apps (Outlook, Edge, Teams, Office). **Why:** BYOD data protection lives in App Protection Policies applied to managed apps; without the apps, BYOD users can't safely reach corporate data. **Verify:** Intune → **Apps** → **App protection policies** can be created; the approved apps are deployed or available. **Implement:** Create App Protection Policies (Implementation §4) and make the approved apps available to BYOD users via Company Portal / store.

R8. Device compliance policy for corporate devices

Description: A compliance policy targeting corporate-owned devices. **Why:** Conditional Access for COD grants access only to **compliant** devices; the policy defines compliant. **Verify:** Intune → **Devices** → **Compliance**; a policy exists and is assigned (scoped to Corporate). **Implement:** Create a compliance policy (OS minimums, encryption, etc.) and target it with the Corporate device filter (Implementation §5).

R9. Device-ownership filters

Description: Intune **device filters** on `deviceOwnership` (Corporate / Personal). **Why:** Filters are how one assignment to a user group applies the *right* policy to the *right* device type — the mechanism that makes "one account, both devices" work cleanly. **Verify:** Intune → **Tenant administration** → **Filters**; filters for Corporate and Personal exist. **Implement:** Create them in Implementation §5, or run the included `Set-OwnershipBasedTargeting.ps1`.

R10. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, app protection check-ins, and authentication depend on these. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level on the marketplace so individual solution listings stay small. Confirm these are in place before deploying *any* Intune solution.

Each entry: **Description · Why required · How to verify · How to implement if missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, enrollment, and security policies. **Verify:** `https://intune.microsoft.com` loads and **Tenant administration** → **Tenant status** shows an active subscription. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies, enrollment connectors, and configuration profiles requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator** (or Global Administrator for connector setup). **Implement:** Assign **Intune Administrator** to the operator; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted devices are enrolled and manageable in Intune. **Why:** Policies only reach enrolled (or app-managed) devices, regardless of platform. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Configure the appropriate enrollment method for the platform and ownership model, then enroll the device.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted to a small group before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists with intended members. **Implement:** Entra → **Groups** → **New group** → type **Security** → add pilot members.

G6. Supported, current operating system

Description: A supported OS baseline with current updates for the targeted platform. **Why:** Modern Intune features and security settings assume a patched, vendor-supported OS. **Verify:** Check the device's OS version or Intune → device → **Hardware**. **Implement:** Deploy update policies or upgrade to a supported release.

G7. Network access to required cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation — Corporate-Owned (COD) & BYOD Device Management

Follow these steps **in order**. The *why* lives in the Solution Overview; the steps below are short and exact so you can **click to implement**. The whole design assumes **one Entra identity per user** that works on both device types — so you assign policies to the **user (group)** and let **ownership-based filters** route the right policy to each device. Pilot first: **Assess → Configure → Pilot → Validate → Expand.**

All steps use the Microsoft Intune admin center at `https://intune.microsoft.com` and Microsoft Entra at `https://entra.microsoft.com` unless noted.

Phase 1 — Identity & enrollment baseline

Step 1. Confirm one identity and licensing per pilot user

1. Entra → **Users** → confirm each pilot user is a **single** account with an Intune + Entra ID P1 license (Requirements R1–R3).

1. Entra → **Groups** → **New group** → **Security** → name it `PILOT – COD-BYOD Users` → add the pilot users.

*(You assign every policy in this solution to this **user** group.)*

Step 2. Set enrollment restrictions for both ownership types

1. **Devices** → **Enrollment** → **Enrollment device platform restrictions.**
2. Open the default (or create a pilot-scoped restriction **named** `Enrollment Restriction – COD-BYOD – Pilot`).
3. For **BYOD** platforms (e.g. iOS/Android personal), set **Personal devices = Allow**.
4. For platforms you only allow as corporate, set **Personal devices = Block** (those must enroll via a corporate method).

Step 3. Establish how COD devices are marked Corporate

1. **Devices** → **Enrollment** → **Corporate device identifiers** → **Add** serial numbers/IMEIs of company devices **(or)** rely on a corporate enrollment program (Autopilot/ADE/Android Enterprise).
1. Confirm a known company device shows **Ownership = Corporate** on its device record.

Phase 2 — Ownership-based targeting (the core of "one account, both devices")

Step 4. Create device-ownership filters

1. **Tenant administration** → **Filters** → **Create.**
2. Filter 1: **Corporate Devices** — platform as needed; rule `(device.deviceOwnership -eq "Corporate")`.
3. Filter 2: **Personal Devices** — rule `(device.deviceOwnership -eq "Personal")`.
4. **Save** both.

> *Prefer automation? Run `Set-OwnershipBasedTargeting.ps1` to create these filters (and optionally assign the policies below) via Microsoft Graph.*

Phase 3 — Corporate-owned (COD): full MDM

Step 5. Create a compliance policy for corporate devices

1. **Devices** → **Compliance** → **Create policy** → pick the platform; **name it** `COD – Compliance – Pilot`.
2. Set baseline rules (OS minimum, disk encryption, firewall, etc.).
3. **Assignments:** assign to **PILOT – COD-BYOD Users**, and under **Filter**, **Include** the **Corporate Devices** filter → **Create**.

Step 6. Create configuration profile(s) for corporate devices

1. **Devices** → **Configuration** → **Create** → **New policy** → platform → your settings; **name it** `COD – Configuration – Pilot`.
2. **Assignments:** assign to **PILOT – COD-BYOD Users**, **Include** the **Corporate Devices** filter.

(Zero-touch corporate enrollment — Windows Autopilot, Apple ADE, Android Enterprise corporate — is each covered by its own solution; plug those in for fully automated COD provisioning.)

Phase 4 — BYOD: App Protection (MAM)

Step 7. Create App Protection Policies

1. **Apps** → **App protection policies** → **Create policy** → **iOS/iPadOS**, **name it** `BYOD – App Protection – iOS – Pilot` (repeat for **Android** as `BYOD – App Protection – Android – Pilot`).
2. **Apps:** select the approved managed apps (Outlook, Edge, Teams, Office, OneDrive).
3. **Data protection:** block **backup**, restrict **cut/copy/paste** to managed apps, block **Save As** to personal locations, require corporate data to stay in managed apps.
1. **Access requirements:** require **PIN** / org credentials to access; set offline timeouts.
2. **Assignments:** assign to **PILOT – COD-BYOD Users**. *(App Protection is identity-based — it applies to the user wherever they sign into the managed app, so it naturally targets the personal device without device enrollment.)*

Step 8. Make the approved apps available to BYOD users

1. **Apps** → **(platform)** → **Add** Outlook, Edge, Teams, Office, Company Portal as needed.
2. Assign them **Available** to **PILOT – COD-BYOD Users** so users install them from Company Portal / store.

Phase 5 — Unify access with Conditional Access

Step 9. One Conditional Access policy for both device types

1. Entra → **Protection** → **Conditional Access** → **Create new policy** → name it `CA – COD-BYOD Access`.
2. **Users: PILOT – COD-BYOD Users**.
3. **Target resources:** Office 365 (and other corporate apps).
4. **Grant:** require **one of the following** — **Require device to be marked as compliant** *(this is satisfied by the corporate device)* **and/or Require app protection policy** *(satisfied by the personal device's managed app)*; set the control to **Require one of the selected controls**.
1. **Enable policy: Report-only** for the pilot → **Create**.

This single policy is what lets the **same account** reach corporate data from a **compliant COD** or an **app-protected BYOD** device.

Phase 6 — Pilot, validate, expand

Step 10. Pilot with one user on two devices

1. As a pilot user, enroll a **corporate** device (it tags Corporate → gets compliance/config).
2. On a **personal** device, sign into **Outlook/Edge** with the **same account** → the App

Protection policy applies (PIN prompt, managed-app data containment).

1. Confirm the **same UPN** now owns both a Corporate and a Personal device in Intune.

Step 11. Validate

1. Run `CODBYOD-Discovery.ps1` (tenant readiness) and `CODBYOD-Validation.ps1 -User <upn>`:

confirm the user has **one Corporate** device (compliant) and **one Personal** device (app protection applied), under a single identity.

1. Confirm the **Corporate** filter applied device policy and the **Personal** filter did not (and

vice-versa for app protection).

1. Review Conditional Access **report-only** results for the pilot user.

Step 12. Expand

1. When the pilot passes, widen the assignments from the pilot group to broader user groups.
2. Move **CA – COD-BYOD Access** from **report-only** to **On**.
3. Roll out in waves, validating each wave with the scripts before widening.

5. Testing & Validation — Corporate-Owned (COD) & BYOD Device Management

Validate against a **pilot user with two devices** (one corporate, one personal) before expanding. Each test has an explicit **pass/fail gate**. Use the included PowerShell scripts (Microsoft Graph) for tenant-side checks and the on-device steps for the user experience.

Test 1 — Tenant readiness

Procedure: Run `CODBYOD-Discovery.ps1`. **Pass:** Enrollment restrictions allow the intended Personal/Corporate split, an App Protection policy exists, a compliance policy exists, Corporate/Personal **device filters** exist, Company Portal is assigned, and a Conditional Access policy is present. **Fail:** Any missing → resolve before piloting.

Test 2 — Single identity, two devices

Procedure: Run `CODBYOD-Validation.ps1 -User <upn>` for the pilot user. **Pass:** The **same UPN** owns at least one **Corporate** device and one **Personal** device. **Fail:** Devices split across different

accounts → consolidate identity (Requirements R1).

Test 3 — Corporate device gets full MDM

Procedure: Inspect the pilot **corporate** device in Intune. **Pass:** Ownership = **Corporate**; the **compliance** and **configuration** policies (Corporate filter) are **applied**; device reports **Compliant**. **Fail:** Policies not applied → confirm the device is tagged Corporate (R5) and the Corporate filter is set on the assignment (Implementation §5–6).

Test 4 — Personal device gets App Protection only

Procedure: On the pilot **personal** device, sign into Outlook/Edge with the pilot account. **Pass:** The App Protection policy applies — **PIN prompt**, corporate data confined to managed apps, **Save As**/paste to personal locations blocked. The device is **not** fully enrolled (or, if personally enrolled, is tagged **Personal** and does **not** receive Corporate device policies). **Fail:** No PIN/containment → confirm the app is approved/managed and the App Protection policy is assigned to the user (Implementation §7).

Test 5 — Ownership filters route correctly

Procedure: Compare policy application across the two devices. **Pass:** The **Corporate** device receives device compliance/config and **not** the BYOD-only items; the **Personal** device receives app protection and **not** corporate device policies. **Fail:** Cross-application → re-check the `deviceOwnership` filter rules and the Include/Exclude on each assignment (Implementation §4).

Test 6 — One account reaches work from both devices

Procedure: With Conditional Access **report-only**, access a corporate web app from the corporate device and a managed app from the personal device, signed in as the same user. **Pass:** Both succeed — the corporate device satisfies **compliant device**; the personal device satisfies **app protection policy** — under the **single** CA policy. **Fail:** One path blocked → review CA grant controls ("require one of"), compliance state, and app protection assignment (Implementation §9).

Test 7 — Unmanaged path is blocked (data protection holds)

Procedure: On the personal device, try to open corporate mail in the **built-in** (unmanaged) mail app or paste corporate data into a personal app. **Pass:** Corporate data is **not** accessible in unmanaged apps; paste/Save-As to personal locations is blocked. **Fail:** Corporate data leaks to unmanaged apps → tighten App Protection data-relocation settings.

Test 8 — Compliance & enforcement

Procedure: Review compliance for the corporate device and the report-only CA results for the user. **Pass:** Corporate device **Compliant**; CA evaluates the user correctly from both devices. **Fail:** Non-compliant or CA misfires → review the compliance policy and CA conditions.

Acceptance gate

Proceed to **Expand** only when Tests 1–8 pass for a pilot user who has **both** a corporate and a personal device signed in with the **same account**, and the data-protection checks (Test 7) hold.

6. Troubleshooting — Corporate-Owned (COD) & BYOD Device Management

Work top-down: confirm the **identity** (one account), then **ownership** (Corporate vs Personal), then **targeting** (filters/assignments), then **access** (Conditional Access). Most issues are a device tagged with the wrong ownership, a missing filter on an assignment, or a BYOD user not using a managed app.

A user's two devices show under different accounts

- **Duplicate identities.** Consolidate to one Entra account; the user must sign in with the same UPN on both devices (Requirements R1). Re-enroll/re-register the stray device under the correct account.

A corporate device is tagged Personal (or vice-versa)

- **Wrong ownership.** Intune → **Devices** → **(device)** → **Properties** → **Device ownership** → set **Corporate**/**Personal**. For COD, add **corporate device identifiers** or use a corporate enrollment program so it tags Corporate automatically (R5).
- After fixing, the device picks up the correct filtered policies on next sync.

Corporate policies are not applying to the corporate device

- **Filter mismatch.** The compliance/config assignment must **Include** the **Corporate Devices** filter and target the **user** group (Implementation §5–6). Verify the filter rule is `(device.deviceOwnership -eq "Corporate")`.
- **Device not compliant/enrolled.** Confirm the device is enrolled and checking in.

App Protection isn't applying to the BYOD device

- **User not in scope.** App Protection assigns to the **user** group — confirm the pilot user is a member (Implementation §7).
- **App not managed/approved.** The user must use an **approved managed app** (Outlook, Edge, Teams).

Built-in mail/browser apps are intentionally excluded.

- **No Intune/MAM license** on the user (R2).
- App Protection applies on **next app launch / token refresh** — close and reopen the managed app.

BYOD user can't access corporate mail/data

- **By design in an unmanaged app.** They must use the managed app (e.g. Outlook), not the built-in

one. Deploy and direct them to the approved apps (Implementation §8).

• **Conditional Access blocking.** If CA requires app protection, accessing from a non-managed app is blocked — expected. Check Entra → **Sign-in logs** for the CA result.

One account works on the corporate device but not the personal one (or vice-versa)

• **CA grant too strict.** The Conditional Access policy must **require *one of*** compliant device **or** app protection — not **both** (Implementation §9). "Require all" blocks one of the two paths.

• **App protection not yet registered** on the personal device — open a managed app and complete the PIN setup.

Personal device received corporate device policies (over-management)

• The personal device was **fully enrolled** and/or tagged **Corporate**. Confirm enrollment restrictions and ownership (R4–R5); for BYOD, prefer **App Protection without enrollment**. If personal MDM enrollment is intended, ensure it's tagged **Personal** so the Personal filter excludes corporate policies.

Conditional Access has unexpected effects

• Keep the pilot policy in **report-only** and read Entra → **Sign-in logs** → **Conditional Access** to see exactly which control was/wasn't satisfied before turning it **On**.

Useful checkpoints

• **Intune:** Devices → (device) → **Properties** (ownership), **Compliance**, **Configuration**; Apps → **App protection policies** → **(policy)** → **Monitor** for per-user status; Tenant administration → **Filters**.

• **Entra: Sign-in logs** → **Conditional Access** tab for grant evaluation.

• **Scripts:** re-run `CODBYOD-Validation.ps1 -User <upn>` after each change to confirm the gate.

7. Best Practices — Corporate-Owned (COD) & BYOD Device Management

Field-proven guidance to keep one identity working cleanly across both device types.

Identity & assignment model

• **Assign policies to user groups, scope with device filters.** This is the heart of the design: one assignment to the user, the right policy follows the device's ownership. Avoid mixing per-device group assignments that fight the filters.

• **Keep one Entra identity per user.** Never create a separate "BYOD account" — it splits policy and breaks Conditional Access logic.

• **Name groups and filters by intent** (e.g. *PILOT – COD-BYOD Users*, *Corporate Devices*, *Personal Devices*) so assignments are self-documenting.

Ownership hygiene

• **Make ownership deterministic.** Use corporate enrollment programs and corporate device identifiers so COD tags **Corporate** automatically; let BYOD default to **Personal**.

• **Audit ownership regularly** with `CODBYOD-DeviceReport.ps1` — a device tagged the wrong way gets the wrong policy set.

BYOD data protection

• **Prefer App Protection (MAM) without enrollment for BYOD.** It protects corporate data without managing the employee's personal device — better adoption, less privacy friction, strong data-leak controls.

• **Lock the data boundary:** block backup, restrict cut/copy/paste and "Save As" to managed apps, require a PIN, and enforce encryption within managed apps.

• **Only enroll a personal device (MDM) when you truly need device settings** — and keep it tagged **Personal** so it never receives corporate device policies.

Corporate device management

• **Apply a real compliance baseline** (OS minimum, encryption, firewall) and gate access on it via Conditional Access.

• **Layer the zero-touch enrollment solutions** (Autopilot / Apple ADE / Android Enterprise) for hands-off COD provisioning; this solution unifies the model they all feed into.

Conditional Access

• **One policy, "require one of."** Grant access when the request is from a **compliant** device **or** an **app-protection-protected** app — never require both, or you lock out one device type.

• **Always start report-only.** Read the sign-in logs for the pilot user before enforcing.

• **Add MFA** as part of the same access story for a complete Zero Trust posture.

Rollout discipline

• **Pilot one user with two real devices** (a corporate and a personal one) — it's the only way to

prove the single-identity experience end to end.

- **Expand in waves**, validating each with `CODBYOD-Validation.ps1` before widening.
- **Communicate the BYOD deal to employees:** what IT can and cannot see/do on a personal device under App Protection (data containment, *not* device surveillance) — clarity drives adoption.

Operations & documentation

- **Keep a runbook** of the pilot/production groups, the two ownership filters, the compliance and app-protection policies, and the Conditional Access policy — the moving parts that define the model.
- **Report monthly** with `CODBYOD-DeviceReport.ps1`: device counts by ownership, users who have both a COD and a BYOD device, compliance, and app-protection coverage.

8. Deliverables Summary — Corporate-Owned (COD) & BYOD Device Management

Everything included in this solution package, and what each piece is for.

Documentation (this guide)

- **Executive Summary** — the business case and outcome in one page.
- **Solution Overview** — the two ownership models, the single-identity design, management models, benefits, and honest limitations.
- **Requirements Guide** — every prerequisite (one identity, MDM+MAM licensing, Entra ID P1, enrollment restrictions, corporate tagging, Company Portal, App Protection, compliance, ownership filters, network) with how to **verify** and how to **implement if missing**, plus the category-level Global Requirements.
- **Implementation Guide** — exact, ordered, click-by-click steps across six phases: identity & enrollment baseline, ownership-based targeting, COD full-MDM, BYOD App Protection, unifying access with Conditional Access, then pilot/validate/expand.
- **Testing & Validation** — eight tests with explicit pass/fail gates and an acceptance gate, proving one account works on both a corporate and a personal device.
- **Troubleshooting** — top-down diagnosis for identity, ownership, targeting, and access failures.
- **Best Practices** — the assign-to-user/scope-by-filter model, ownership hygiene, BYOD data protection, COD management, Conditional Access, and rollout discipline.

PowerShell scripts (Microsoft Graph) — `CODBYOD-Scripts.zip`

All scripts are tenant-side (run from an admin workstation with the Microsoft Graph PowerShell SDK); each has a header explaining **Purpose · How to run · Expected results**.

1. `CODBYOD-Discovery.ps1` — tenant readiness: enrollment restrictions (personal allowed?), compliance policies, App Protection policies, Corporate/Personal device filters, Company Portal, and Conditional Access presence.
1. `CODBYOD-Validation.ps1` — for a given user, confirms a single identity owns both a **Corporate** device (compliant) and a **Personal** device (app protection applied), with a PASS/FAIL summary.
1. `CODBYOD-DeviceReport.ps1` — tenant-wide CSV: devices by **ownership**, platform, enrollment type, compliance, and primary user — plus a list of users who have **both** a COD and a BYOD device.
1. `CODBYOD-Remediation.ps1` — safe remediation: set device **ownership**, re-assign policies to the target group, and flag enrollment-restriction/filter misconfigurations. Read-only by default; changes require an explicit confirmation switch.
1. `Set-OwnershipBasedTargeting.ps1` — creates the **Corporate** and **Personal** `deviceOwnership` filters (and optionally assigns the compliance/app-protection policies with those filters) — the mechanism that makes one account work cleanly across both device types.

Hero image

- `CODBYOD-hero.png` — the branded product image (1200×1200) used on the marketplace listing.

How it's delivered

A branded **Word** document and **PDF** (cover, table of contents, page numbers, and a scripts appendix), plus all five PowerShell scripts in a single **ZIP** — yours to keep and reuse across **unlimited devices**.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

CODBYOD-DeviceReport.ps1

Purpose: Exports every managed device with ownership (Corporate/Personal), platform, enrollment type, compliance, and primary user — then lists users who own BOTH a corporate and a personal device (the population this solution is designed for).

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\CODBYOD-DeviceReport.ps1 .\CODBYOD-DeviceReport.ps1 -CsvPath C:\Temp\cod-byod.csv

CODBYOD-Discovery.ps1

Purpose: Confirms the building blocks of the one-identity / two-ownership model exist: enrollment restrictions, a compliance policy (for COD), App Protection policies (for BYOD), the Corporate/Personal device filters, Company Portal, and a Conditional Access policy.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\CODBYOD-Discovery.ps1 Reads only — makes no changes.

CODBYOD-Remediation.ps1

Purpose: The common fixes that unblock the COD/BYOD model: correct a device's ownership (Corporate/Personal) so the right filtered policies apply, and audit for devices whose ownership is Unknown. Read-only by default.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once # Audit only (no changes): .\CODBYOD-Remediation.ps1 # Set a device's ownership (explicit opt-in): .\CODBYOD-Remediation.ps1 -Apply -DeviceName "Jane-Surface" -Ownership company

CODBYOD-Validation.ps1

Purpose: For a given user (UPN), lists their Intune devices and validates the acceptance gate: the same account owns at least one Corporate device (compliant) and at least one Personal device, and an App Protection policy targets the user. Produces a clear PASS/FAIL verdict.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\CODBYOD-Validation.ps1 -User jane@contoso.com

Set-OwnershipBasedTargeting.ps1

Purpose: Creates the two Intune assignment filters on `deviceOwnership` that make "one account, both device types" work: a Corporate filter (routes full-MDM policy to company devices) and a Personal filter (keeps corporate device policy off BYOD). Idempotent — skips a filter that already exists.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\Set-OwnershipBasedTargeting.ps1 -WhatIf # preview .\Set-OwnershipBasedTargeting.ps1 # create